

Pinout / wiring section for assembly guide

Pico Spool Monitor Pinout

The spool monitor uses one GPIO input and one GPIO output per lane.

Switch inputs:

- Each lane switch connects between its assigned GPIO pin and GND.
- The firmware uses the Pico's internal pull-up resistors.
- Switch closed = filament/spool present.
- Switch open = filament/spool empty.

LED outputs:

- Each lane LED connects from its assigned GPIO pin through a resistor to the LED anode.
- LED cathode connects to GND.
- Recommended resistor: 220Ω – 330Ω .

Power:

- The Pico 2 W is powered through its USB port.
- All switch grounds and LED grounds must connect to Pico GND.

4-lane pinout

Lane 1:

- Switch signal: GP2
- LED signal: GP6

Lane 2:

- Switch signal: GP3

- LED signal: GP7

Lane 3:

- Switch signal: GP4

- LED signal: GP8

Lane 4:

- Switch signal: GP5

- LED signal: GP9

8-lane pinout

Lane 1:

- Switch signal: GP2

- LED signal: GP6

Lane 2:

- Switch signal: GP3

- LED signal: GP7

Lane 3:

- Switch signal: GP4

- LED signal: GP8

Lane 4:

- Switch signal: GP5

- LED signal: GP9

Lane 5:

- Switch signal: GP10
- LED signal: GP14

Lane 6:

- Switch signal: GP11
- LED signal: GP15

Lane 7:

- Switch signal: GP12
- LED signal: GP16

Lane 8:

- Switch signal: GP13
- LED signal: GP17

Wiring table

Lane	Switch GPIO	LED GPIO
1	GP2	GP6
2	GP3	GP7
3	GP4	GP8
4	GP5	GP9
5	GP10	GP14
6	GP11	GP15
7	GP12	GP16
8	GP13	GP17

Switch wiring

Each switch should be wired like this:

Pico GPIO ---- switch ---- GND

Example for Lane 1:

GP2 ---- Lane 1 switch ---- GND

Because the firmware uses pull-up inputs, no external pull-up resistor is required.

LED wiring

Each LED should be wired like this:

Pico GPIO ---- resistor ---- LED anode

LED cathode ---- GND

Example for Lane 1:

GP6 ---- 220 Ω resistor ---- LED anode

LED cathode ---- GND

Ground note

All grounds are shared. If the PCB has many switch and LED grounds, use a common ground rail or ground bus on the prototype board.

Pico GND

└— switch grounds

└— LED cathodes

└— any connector grounds

LED behavior

Loaded and stable:

- LED off

Recently loaded/recovered:

- LED solid on briefly

Empty active lane:

- LED blinking

Disabled/inactive lane:

- LED off

1. Print the parts

2. Install the PCB into the base using M2 screws

3. Route front-side wires through the front exit

4. Route rear-side wires through the rear exit

5. Secure wire bundles with zip ties through the strain relief bridges

6. Confirm Pico USB cable clearance

7. Attach lid using M2.5 screws

8. For extrusion mounting, attach adapter plate using M2.5 x 6 mm screws

9. Attach adapter plate to 2020/4040 extrusion using M5 screws and T-nuts
10. Power on and verify Pico dashboard / lane LEDs